2017 Summer Qual

III.

$$\hat{y}_k = \hat{\beta}_0 + \hat{\beta}_1 x_k = \bar{y} + \hat{\beta}_1 (x_k - \bar{x})$$

$$= \sum_{j=1}^{n} \left(\frac{1}{n} + \frac{(x_k - \bar{x})(x_j - \bar{x})}{S_{xx}} \right) y_j$$

$$\mathbb{E}[y_k] = \beta_0 + \beta_1 x_k$$

$$\mathrm{var}(\hat{y}_k) = \sum_{j=1}^{n} \left(\frac{1}{n} + \frac{(x_k - \bar{x})(x_j - \bar{x})}{S_{xx}} \right)^2 \sigma^2$$

$$= \sigma^2 \left(\frac{1}{n} + \frac{(x_k - \bar{x})^2}{S_{xx}} \right)$$

$$\frac{\hat{y}_k - \mathbb{E}(y_k)}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{(x_k - \bar{x})^2}{S_{xx}} \right)}} \sim N(0,1)$$

$$\frac{\hat{\sigma}^2}{\sigma^2} = \frac{SSE}{\sigma^2} \Big/ n-1 \sim \chi^2(n-1) \Big/ n-1$$

$$\frac{\hat{y}_k - \mathbb{E}(y_k)}{\sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(x_k - \bar{x})^2}{S_{xx}} \right)}} = \frac{\dfrac{\hat{y}_k - \mathbb{E}(y_k)}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{(x_k - \bar{x})^2}{S_{xx}} \right)}} \sim N(0,1)}{\sqrt{\dfrac{\hat{\sigma}^2}{\sigma^2}} \sim \chi^2(n-1)/n-1} \quad \perp\!\!\!\perp$$

$$\left(\hat{\beta} \perp\!\!\!\perp \hat{\sigma}^2 \because \sigma^2 (X^T X)^{-1} X^T (I - H_X) = 0 \right) \sim t(n-1)$$

$\boxtimes$

$\boxed{2}$.

$$R^2 = \frac{SSR}{SST} = \frac{SSR}{SSR+SSE} \quad \& \quad \begin{aligned}&(H_X - H_1)(I - H_X)\\ &= H_X - H_X - H_1 + H_1\\ &= 0\end{aligned}$$

$$= \frac{\frac{SSR}{\sigma^2}}{\frac{SSR}{\sigma^2} + \frac{SSE}{\sigma^2}} = \frac{\chi^2(p)}{\chi^2(p) + \chi^2(n-p-1)}$$

$$= \frac{\Gamma(\frac{p}{2}, 2)}{\Gamma(\frac{p}{2}, 2) + \Gamma(\frac{n-p-1}{2}, 2)} \sim \text{Beta}\left(\frac{p}{2}, \frac{n-p-1}{2}\right)$$

$\boxtimes$

$\boxed{3}$.

$$A\beta = 0, \quad AC = 0 \Rightarrow \beta = C\theta \quad \because \beta \in N(A)$$

$$\min_{\beta} (y - X\beta)^T (y - X\beta) + \lambda(A\beta - 0)$$

$$\Longleftrightarrow \min_{\theta} (y - XC\theta)^T (y - XC\theta) = L$$

$$\frac{\partial L}{\partial \theta} = -2C^T X^T (y - XC\theta) = 0$$

$$\Rightarrow C^T X^T y - (C^T X^T X C)\theta = 0$$

$$\Rightarrow \hat{\theta} = (C^T X^T X C)^{-1} C^T X^T y$$

full column rank, full row rank $\Rightarrow C^T X^T X C$ is nonsingular

$$\Rightarrow \hat{\beta} = C(C^T X^T X C)^{-1} C^T X^T Y$$

$\boxtimes$